Review

1) Find the average value of the function on the given intervals

$$f_{ave} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

a) $f(x) = \dfrac{x}{\sqrt{9+x^2}}$, $[1,9]$

$$f_{ave} = \frac{1}{9-1}\int_1^9 \frac{x}{\sqrt{9+x^2}}\,dx \quad , \quad u = 9+x^2, \quad du = 2x\,dx$$

$$= \frac{1}{8}\int \frac{\cancel{x}}{\sqrt{u}}\frac{du}{2\cancel{x}} = \frac{1}{8}\int \frac{1}{2\sqrt{u}}$$

$$= \frac{1}{8}\int \frac{1}{2\sqrt{u}}\,du = \frac{1}{8}(\sqrt{u})$$

$$= \frac{1}{8}\left(\sqrt{9+x^2}\right)\Big|_1^9 = \frac{1}{8}(\sqrt{90} - \sqrt{10})$$

$$= \frac{1}{8}(3\sqrt{10} - \sqrt{10}) = \frac{\sqrt{10}}{4}$$

b) $g(x) = e^{\tan(x)}\sec^2(x)$, $[0, \pi/4]$

$$g_{ave} = \frac{1}{\pi/4 - 0}\int_0^{\pi/4} e^{\tan(x)}\sec^2(x) = \frac{4}{\pi}\int_0^{\pi/2} e^{\tan(x)}\sec^2(x)$$

let $u = \tan x$ $\Rightarrow$
$du = \sec^2 x\,dx$

$$= \frac{4}{\pi}\int e^u \sec^2(x)\frac{du}{\sec^2 x}$$

$$= \frac{4}{\pi}\int e^u\,du = \frac{2}{\pi}(e^u)\Big|$$

Substitute $\tan x$ back

$$= \frac{4}{\pi}e^{\tan x}\Big|_0^{\pi/4} = \frac{4}{\pi}(e^1 - e^0) = \frac{4}{\pi}(e-1)$$

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c) $h(x) = 5xe^{-x^2}$, $[5, 10]$

$h_{ave}: \frac{1}{10-5}\int_5^{10} 5xe^{-x^2}\,dx = \int_5^{10} xe^{-x^2}\,dx$

let $u = -x^2$

$du = -2x\,dx \Rightarrow dx = \frac{-du}{2x}$

$\int xe^{u}\frac{du}{-2x} = -\frac{1}{2}\int e^{u}\,du = -\frac{1}{2}e^{u}|$

Substitue $-x^2$ back.

$-\frac{1}{2}(e^{-x^2})\Big|_5^{10} = -\frac{1}{2}(e^{-100} - e^{-25})$

Evaluate the integral

a) $\int x^2\cos(x)\,dx$

$u = x^2$ $\quad dv = \cos x$

$du = 2x$ $\quad v = \sin x$

$\int x^2\cos(x) = x^2\sin x - \int 2x(\sin x)$

$2\int x\sin x \Rightarrow u = x \quad dv = \sin x$

$du = 1 \quad v = -\cos(x)$

$\int x\sin x = -x\cos(x) - \int(1)(-\cos x)$

$2\int x\sin x = 2(-x\cos(x) + \sin x)$

$\int x^2\cos(x) = x^2\sin x - (2(-x\cos(x) + \sin x))$

$= x^2\sin x + 2x\cos(x) - 2\sin x + C$

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b) $\int \arcsin(2x)\,dx$

let $u = \arcsin(2x)$, $dv = dx$

$du = \frac{2}{\sqrt{1-(2x)^2}}$ $\quad v = x$

$$\int \arcsin(2x) = x\arcsin(2x) - \int \frac{2x}{\sqrt{1+(2x)^2}}$$

$$\int \frac{2x}{\sqrt{1+4x^2}} = -\frac{1}{2}\sqrt{u} = -\frac{1}{2}\sqrt{1-4x^2}$$ ---- using u-sub with $u = \sqrt{1-4x^2}$

$$\int \arcsin(2x) = x\arcsin(2x) + \frac{\sqrt{1-4x^2}}{2} + C$$

c) $\int t^2 5^{-2t}\,dt$

$u = t^2$ $\quad dv = 5^{-2t}\,dt$

$du = 2t\,dt$ $\quad v = -\frac{1}{2\ln 5} 5^{-2t}$

$$= \frac{-t^2}{2\ln 5} 5^{-2t} - \int -\frac{1}{2\ln 5} 5^{-2t}(2t)\,dt = \frac{-t^2 5^{-2t}}{2\ln 5} + \frac{1}{\ln 5}\int t\, 5^{-2t}\,dt$$

$u = t$ $\quad dv = 5^{-2t}\,dt$

$du = dt$ $\quad v = \frac{5^{-2t}}{-2\ln 5}$

$$= \frac{-t^2 5^{-2t}}{2\ln(5)} + (\ln(5))^{-1}\left(\frac{t\,5^{-2t}}{-2\ln 5} - \int \frac{-1}{2\ln 5} 5^{-2t}\,dt\right)$$

$$= \boxed{\frac{-t^2 5^{-2t}}{2\ln(5)} - \frac{t\,5^{-2t}}{2(\ln 5)^2} - \frac{1}{4(\ln 5)^3} 5^{-2t} + C}$$

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$$\int \tan^{-1}\left(\frac{1}{x}\right) dx$$

$u = \tan^{-1}(1/x)$ $\qquad dv = dx$

$du = \dfrac{1}{1+\left(\frac{1}{x}\right)^2} \cdot \left(\frac{-1}{x^2}\right)$ $\qquad v = x$

$$= x\tan^{-1}(1/x) - \int x\left(\frac{1}{1+1/x^2} \cdot \left(\frac{-1}{x^2}\right)\right) dx$$

$$= x\tan^{-1}(1/x) - \int \frac{-x}{x^2+1}$$

$, u = x^2+1$

$du = -2x\,dx$

$$= \boxed{x\tan^{-1}\left(\frac{1}{x}\right) + \frac{1}{2}\ln|x^2+1| + C}$$

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